

ZAM AI

The AI Health Assistant for Africa and the World

Master Working Document · Confidential

1. Vision & Mission

Zam AI is the intelligent health assistant designed to power better medical decisions for patients, pharmacists, doctors, and healthcare systems worldwide.

Its mission is to combine artificial intelligence, trusted medical knowledge, and real-world healthcare data to improve:

- Patient safety
- Access to healthcare, especially in underserved areas
- Medical decision-making speed and accuracy
- Health system efficiency

2. Core Architecture

Zam AI is built on five layered components that work together to deliver safe, intelligent, and contextual health responses.

Layer 1 — Medical Knowledge Layer

The foundation of all AI responses. Pulls from verified, authoritative medical references:

- NAFDAC
- EMDEX (Nigerian Drug Index)
- BNF — British National Formulary
- MIMS (used widely across African and Asian markets)
- NEML — Nigeria Essential Medicines List
- WHO ATC Classification CSV (global drug taxonomy)
- ZamdaHealth live database (pharmacy inventory, prescription history, patient records)

Layer 2 — Pharmaceutical Intelligence Layer

Drug-specific intelligence: dosage guidance, interactions, contraindications, availability, and alternatives. The WHO ATC codes power the drug categorisation and alternative suggestion engine.

Layer 3 — Patient Health Layer

Patient-specific context: symptoms, prescription history, medication adherence, and health risk profile. This layer personalises responses per individual.

Layer 4 — Predictive AI Layer

Machine learning models that run in the background to analyse disease patterns, drug demand, patient health risks, and outbreak signals across anonymised aggregate data.

Layer 5 — Conversational AI Layer (Voice + Text)

The interface users interact with. Supports:

- Natural language text chat
- Multilingual voice input and output — English, Pidgin, Hausa, Yoruba, Igbo
- Auto language detection — system detects the language spoken and responds in the same language

Critical Constraint — Hallucination Prevention

All AI medical responses must be generated using RAG (Retrieval-Augmented Generation) anchored to the verified data sources above. The model must NEVER answer medical questions from its general training knowledge alone. This is non-negotiable.

3. User Stories — Patients (Phase 1)

Patients are the primary Phase 1 focus. The goal is to give any person in Nigeria — regardless of literacy level or language — access to reliable health guidance 24/7.

SYMPTOM CHECKING & TRIAGE

US-P01

As a **patient**

I want to type or speak my symptoms in English, Pidgin, Hausa, Yoruba, or Igbo so that I can get a clear response about what might be wrong and what I should do — even if I am not comfortable typing.

US-P02

As a **patient**

I want to receive a suggestion for an appropriate over-the-counter medication for mild conditions like malaria symptoms, headache, or fever

So that I can manage common illnesses without an unnecessary hospital visit.

US-P03

As a **patient**

I want to receive a warning when my described symptoms suggest something serious

So that I know when to see a doctor urgently instead of self-medicating.

PRESCRIPTION & MEDICATION SUPPORT

US-P04

As a **patient**

I want to upload or scan a prescription and receive a plain-language explanation of each drug — what it is, how to take it, side effects, and what to avoid

So that I understand my treatment clearly without having to ask a pharmacist or doctor to repeat themselves.

US-P05

As a **patient**

I want to receive push notifications reminding me when to take my medication and when to refill my prescription

So that I stay on track with my treatment and do not miss doses.

US-P06

As a **patient**

I want to ask Zam AI questions about my medication in my own language at any time of the day

So that I get accurate answers without waiting for a pharmacy to open.

PROACTIVE HEALTH MONITORING

US-P07

As a **patient**

I want to receive seasonal illness warnings based on my location and the current disease pattern in my area

So that I can take preventive action before I fall sick.

US-P08

As a **patient**

I want to view a personal health report summarising my medication history, adherence record, and upcoming health risks
So that I have a clear picture of my health over time.

4. User Stories — Pharmacies (Phase 2)

Pharmacists need speed, safety, and business intelligence. Zam AI plugs into the existing POS system and provides real-time clinical and business support at the point of dispensing.

PATIENT SAFETY AT POINT OF SALE

US-PH01

As a **pharmacist**

I want to see an automatic real-time alert when a drug I am about to dispense has a dangerous interaction with something already in the patient's prescription history
So that I catch dangerous combinations before they leave the counter — protecting the patient and the pharmacy.

US-PH02

As a **pharmacist**

I want to receive a suggested alternative drug when the prescribed medicine is out of stock
So that I can serve the patient immediately with a bioequivalent substitute instead of turning them away.

US-PH03

As a **pharmacist**

I want to receive a flag when a dosage I am entering looks unusual or unsafe based on the patient's age, weight, or condition
So that I can verify before dispensing and prevent accidental overdose or underdose.

US-PH04

As a **pharmacist**

I want to generate a clear printed or digital medication instruction sheet for the patient automatically
So that Patients receive proper guidance with every purchase without extra effort from me.

US-PH05

As a **pharmacist**

I want to ask Zam AI a drug question verbally while my hands are occupied at the counter

So that I get the information I need immediately without stopping what I am doing.

BUSINESS INTELLIGENCE

US-PH06

As a **pharmacy owner**

I want to see a weekly demand forecast per drug category based on season, region, and disease patterns

So that I can reorder intelligently before stock runs out rather than reacting after.

US-PH07

As a **pharmacy owner**

I want to receive automated smart reorder suggestions based on my sales history and current forecasts

So that I spend less time on procurement decisions and reduce both stockouts and overstock.

US-PH08

As a **pharmacy owner**

I want to receive alerts for products likely to expire before they are sold

So that I take action early — discount, return, or transfer — and reduce waste.

US-PH09

As a **pharmacy owner**

I want to see profitability insights identifying my high-volume low-margin products

So that I can renegotiate supplier terms or adjust pricing to improve margins.

US-PH10

As a **pharmacy owner**

I want to compare my pharmacy's performance against anonymised regional benchmarks

So that I understand how I am performing relative to similar pharmacies and where to improve.

5. User Stories — Health Professionals (Phase 2)

Doctors, consultants, and clinical pharmacists need fast, evidence-based clinical support. Zam AI acts as a second pair of eyes during consultations and ongoing patient management.

CLINICAL DECISION SUPPORT

US-D01

As a **doctor**

I want to enter a patient's current medications and a proposed new prescription, and receive a validation check against drug interaction databases and treatment guidelines
So that I can prescribe with confidence and catch dangerous combinations before they reach the patient.

US-D02

As a **doctor**

I want to run a symptom analysis during a consultation by describing what the patient presents with
So that I get AI-supported differential diagnosis points to consider alongside my clinical judgement.

US-D03

As a **doctor**

I want to receive treatment guideline suggestions aligned to NEML, BNF, and local Nigerian standards for a given condition
So that My prescribing stays evidence-based and within approved national guidelines.

US-D04

As a **doctor**

I want to upload a patient's lab results and receive an AI-generated plain-language interpretation summary
So that I can move through consultations faster, especially during high-volume clinic days.

US-D05

As a **doctor**

I want to dictate a patient's symptoms during a consultation and have Zam AI run a real-time background check
So that I keep the consultation flowing naturally without stopping to type queries manually.

PATIENT MANAGEMENT

US-D06

As a **doctor**

I want to set follow-up reminders for patients with chronic conditions and see whether they have collected their medication refills

So that I can monitor adherence and flag patients who may be deteriorating without coming back.

US-D07

As a **doctor**

I want to receive automated alerts when a patient I am managing shows risk signals based on their health data and purchase history

So that I can intervene proactively rather than waiting for an emergency visit.

6. Zam AI as a SaaS API — Third-Party Integration

Beyond internal ZamdaHealth usage, Zam AI is designed to be sold as a standalone medical intelligence API to other healthtech companies across Africa.

US-A01

As a **third-party healthtech company**

I want to call the Zam AI API to run drug interaction checks, symptom analysis, and dosage verification within my own application

So that I can add medical intelligence to my product without building and maintaining my own AI health model.

US-A02

As a **API consumer**

I want to authenticate via API key and receive all responses in a structured, well-documented JSON format

So that I can integrate Zam AI into any frontend or backend stack quickly and reliably.

US-A03

As a **API consumer**

I want to access the Zam AI SaaS dashboard as a separate interface from ZamdaHealth's internal admin

So that My data and usage metrics are isolated and manageable within my own account.